



Climate proxies for the last 17.3 ka from Lake Hazar (Eastern Anatolia), extracted by independent component analysis of μ -XRF data



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ABSTRACT

The elemental composition of lake sediment cores is often the result of several independent processes. In this study we attempt to extract statistically independent climate related signals from μ -XRF multi element data of a core drilled from Lake Hazar in Eastern Anatolia, using the independent component analysis (ICA) method. In addition, we analysed ostracod shells for oxygen and carbon isotopes. The ICA method has advantages over traditional dimension reduction methods, such as principal component analysis or factor analysis, because it is based on maximal statistical independence rather than uncorrelatedness, where independence is a stronger property.

The Hz11-P03 core, which represents the last 17.3 ka, was recovered from Lake Hazar which, at times, formed the headwaters of the Tigris. Applying the ICA method, we selected two out of six independent components by measuring distance correlation similarity. We propose that one of the selected components can be read as a proxy for temperature and the other for precipitation in this region.

Our results indicate that the region was relatively cold and wet during the late glacial, between 17.3 and 14.8 ka BP, and wet and warm during Bölling-Allerød. The lake level dropped below today's level during the Younger Dryas stadial (12.49 and 11.76 ka BP), forming a marked hiatus in the core's stratigraphic record. During the beginning of the Holocene, while precipitation values were high, the temperature gradually increased until 8 ka BP. Between 8 and 5 ka BP, the region was warm but extremely dry. After 5 ka BP, around 3.5 ka BP temperatures suddenly fell, and three abrupt dry phases are observed around 3.5 ka and 2.8 ka and 1.8 ka BP.

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1. Introduction

Geochemical proxies are the most widely used indicators of Quaternary paleoclimates. In particular, the geochemical "archives" contained in lacustrine sediments provide long and high-resolution proxies of terrestrial records. However, such geochemical proxies, especially those gathered from the lakes, can only be indirect indicators of past climates (Roberts et al., 2008) and may be influenced, to different degrees, by several independent factors. In order to overcome this indirectness, multi-proxy methods are widely

used. However, it may be complicated and misleading to interpret multi-proxy data qualitatively in order to form a single composite impression about the past dynamics that govern those proxies. Part of this shortcoming is due to the fact that, commonly used data reduction techniques, such as principal component analysis (PCA) and factor analysis (FA), are not capable of imposing certain probabilistic constraints on the problem exposed by the multi-proxy data. The orthogonal transformation of PCA and FA point out the directions of maximal variance. However, maximal variance may not be the researcher's main aim, unless it is aimed to find for the orthogonal directions ordered in maximal changes through the data. On the other hand, the resultant sediment geochemistry can be assumed to be a linear combination of statistically independent processes. In this study, instead of the widely used data reduction methods which depend on uncorelatedness (here by

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uncorrelatedness of two random variables, as defined in probability theory, two random variables are uncorrelated, if their covariance is equal to zero) of the resulting vectors, we apply the method of independent component analysis (ICA) with the aim of extracting the statistical independent/quasi-independent processes of the past climate causing the variability in the proxy data sets.

Micro-X-ray fluorescence (μ -XRF) of elemental profiles can be used as proxies for different processes in lakes with differing environmental settings (cf. Davies et al. (2015)). The statistical dependence of μ -XRF elemental profiles of a core is widely recognized by the research community. For example, instead of elemental profiles, it is a standard practice to use elemental ratio profiles to normalize the effects of some process-related factors and possible systematic errors (cf. Croudace and Rothwell (2015)). The elemental profiles of lake cores are the results of physical/statistical independent or quasi-independent processes and parameters, such as precipitation, organic production, temperature, water level and redox conditions in the lake or in its drainage basin.

In previous studies, PCA and FA are the most widely used statistical transformation methods applied to extract the underlying components of proxy data. PCA and FA give useful results if the observed time series are normally distributed and are linear mixtures of the source series (Aires et al., 2000). However, even modern climate variables rarely obey Gaussian distribution (Perron and Sura, 2012; Sura and Hannachi, 2014), and shifts, trends and abrupt events increase the degree of past climate variables' non-Gaussianity and non-linearity. It is unreasonable to expect the data to be in the form of linear mixtures when one considers the highly dynamic environments from which they are sampled. The FA method does not actually extract components, rather it explains the governing factors of the system and the constituents of each factor, in the second order statistical sense. The components extracted by PCA are uncorrelated and the method tries to find the components that explain the maximal variance. Since the extracted components with maximum variances are not separated sources, instead they are mixtures of the true signals (Stone, 2004), the objective of PCA is questionable in the sense of source separation. This low-order nature is due to the fact that the backbone of both FA and PCA methodologies are the variance-covariance matrices.

ICA is a more robust statistical transformation method, developed for signal separation, and provides an alternative to PCA (Comon, 1994). ICA's fundamental assumption is that the source time series are statistically independent for each point in time (Hyvärinen and Oja, 2000) and aims to identify a rotation matrix that in practice maximizes the independence of each source component. The main difference between the aims of PCA/FA and ICA is that the former aims to find uncorrelated components, whereas the latter aims to extract independent ones. Since, independency of random variables implies uncorrelatedness, the assumption of being independent is more powerful than being uncorrelated.

ICA has proven its explanatory power in different subjects (see reviews of Acharya and Panda (2008) and Naik and Kumar (2011)), such as feature extraction, signal processing, image processing, telecommunications, financial time series, etc. There are also, some examples in climate research which use ICA as an exploratory tool. By using ICA, Aires et al. (2000) claim to reveal the geographical variability of El Niño Southern Oscillation and its links with Atlantic Ocean. Basak et al. (2004), Mori et al. (2006) and Hannachi et al. (2009) extracted teleconnection patterns from sea level pressure field via ICA. Tatlı et al. (2004, 2005) offer ICA as a preprocessing method in order to reduce dimensions through the downscaling process of general circulation model outputs. However, ICA has never been applied to paleoclimate time series. Several different

approaches to ICA exist, such as those based on the infomax or maximum likelihood methods (for an extended review see Choi et al. (2005)). We use the FastICA algorithm (Hyvärinen and Oja, 2000), because it is a rapid and computationally straightforward method.

In this study, by applying FastICA, we attempt to extract precipitation and temperature related components from the μ -XRF data of a 3.5 m long piston core (Hz11-P03) recovered from Lake Hazar (Eriş, 2013; Eriş et al., submitted). Hz11-P03 spans the last 17.3 ka, allowing us to understand past climate variability for the region throughout the last glacial-interglacial transition and the Holocene. Lake Hazar lies in the Eastern Anatolia region, feeds the headwaters of the major rivers of the Mesopotamia. Therefore, this study aims to give not only the climate variability of Lake Hazar and its catchment basin, also aims to give insight to the water variability in the so-called cradle of civilization.

There are some past climate studies in the region, such as geochemical and paleoecological data from Lake Nar and Eski Acıgöl (Roberts et al., 2001, 2016; Woldring and Bottema, 2003; Jones et al., 2006; Turner et al., 2008; Dean et al., 2015). There are also longer terrestrial records from Lake Van, which span the last 600 ka BP (see, Litt and Anselmetti (2014)). However, none of the aforementioned studies use statistical decomposition methods, except Stockhecke et al. (2016), which use PCA of Lake Van geochemistry data to reveal hydroclimate variability in Eastern Mediterranean for the last 360 ka BP.

2. Regional setting

Lake Hazar is located in south-eastern Anatolia at an altitude of 1255 m (Fig. 1). It is a 25 km long, 7 km wide intra-montane sedimentary basin with a surface area of 78.5 km² and a volume of 7.5×10^9 km³. The maximum depth of the lake is 220 m. Lake Hazar is a tectonic lake, located in an active pull-apart basin along the East Anatolian Fault Zone (Hempton and Dunne, 1984; Garcia Moreno et al., 2011). Eastern Anatolia is the main water source for the major Mesopotamian rivers, Tigris and Euphrates. Lake Hazar lies on the upstream of a tributary of the Tigris river, which was previously an outlet of the lake (Çetin et al., 2003; Nicoll, 2010). Tectonic uplift of the south-eastern part of the lake and excessive use of for agricultural purposes has broken the connection between the lake and the Tigris. Today, it is a terminal lake and fed only by numerous small ephemeral streams.

The main water and sediment sources for the lake are the Kürkçay and Zikkım rivers which flow into the south-west and north-east of the lake, respectively. Maximum discharge occurs during the spring season as a result of snow melt and rainfall. The lake's catchment area is mostly covered by magmatic, metamorphic and ophiolitic rocks of the Mesozoic and Paleozoic Eras. The main bedrock along the southern coast of the lake, its most rugged area, also consists of Middle Eocene calcareous rocks and Late Jurassic magmatic rocks.

Lake Hazar is a monomictic, oligotrophic and alkaline lake (pH = 9.1 ± 0.2) with high carbonate and bicarbonate concentrations. Complete mixing takes place during the autumn and early winter and incomplete mixing in the spring season. The lake is stratified between June and September, and the surface water temperature changes from 5 °C to 29 °C in the pelagic zone throughout the year (Koçer and Şen, 2012).

The climate of Eastern Anatolia has long been considered as being a transition between continental and Mediterranean climates and is described as a Continental Mediterranean climate (Türkeş and Tatlı, 2011) which is distinguished by excess precipitation in spring, rather than in winter (Erol, 2011). Precipitation in the

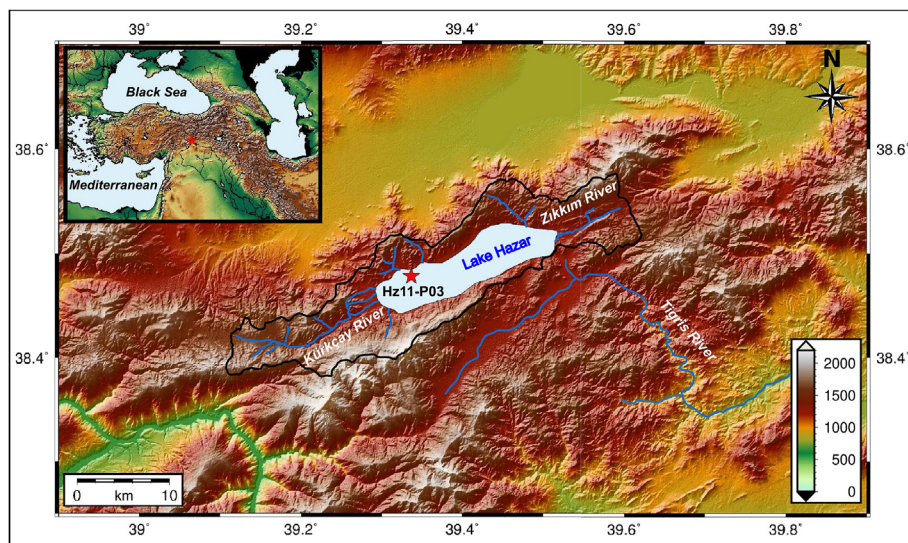


Fig. 1. Location map showing the drainage basin of Lake Hazar. The star indicates the location of the Hz11-P03 piston core. This map was prepared using the free mapping tool GMT, version 5.2.2 (Wessel et al., 2013) and the catchment area and rivers were plotted using the free GIS tool SAGA, version 2.1.4 (Conrad et al., 2015).

Mediterranean basin in winter is mainly affected by the position and power of the subtropical anticyclone settled over the Azores. Weaker high pressure over the Azores is one of the main causes of wetter winters (Lionello et al., 2006a). However, Eastern Mediterranean precipitation is also fed by moisture from the western Mediterranean (Fernández et al., 2003) and the Siberian high pressure system has an effect on winter precipitation and temperatures (Lionello et al., 2006b). The northward movement of the subtropical high pressure system in the summer leaves the region almost with no precipitation at all. Furthermore, the thermal low, a result of the south Asian summer monsoon, that settles over the Persian Gulf carries dry air in from the Asian interior (Bozkurt et al., 2012).

The lake area has an annual mean precipitation of 610.1 mm for the time period 1974–2004, according to the closest meteorology station to the study area which is at Sivrice in Elazığ. This precipitation falls mainly between November and May, accounting for more than 87% of the total precipitation.

3. Materials and methods

3.1. Core Hz11-P03

The 3.8 m long core, Hz11-P03, was recovered in 2009 from north-west of Lake Hazar, at a depth of 54 m (Fig. 1). The physical, sedimentological and geochemical methods applied to understand the properties of this core through various analyses are described in detail by Eriş et al. (submitted).

3.2. μ -XRF core scanner analysis

The split half core was non-destructively analysed at 1 mm resolution, using an ITRAX XRF core scanner equipped with a Mo X-ray source set to 60 kV, 0.3 mA and an exposure time of 30 s at the EMCOL Core Analysis Laboratory at İstanbul Technical University. Up to 25 elements are semi-quantitatively analysed and concentrations were recorded as counts per second (cps).

3.3. Magnetic susceptibility measurements

Magnetic susceptibility (MS) measurements were performed in İTÜ-EMCOL laboratories via Geotek Multi-Sensor Core Logger at 1 cm resolution.

3.4. $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ analysis

The split half core was sampled at 30 mm intervals for stable oxygen and carbon analysis from ostracod shells. The samples were freeze-dried and sieved through 63 μm mesh under filtered tap water. Ostracod species were identified under a stereo microscope and the most continuous and abundant species, *Candona neglecta* (Sars, 1887), were selected for the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ analysis. For each section 20 adult carapace samples were analysed at the University of Arizona stable isotope and geochemistry laboratories, in the USA.

$\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ of carbonates were measured using an automated carbonate preparation device (KIEL-III) coupled to a gas-ratio mass spectrometer (Finnigan MAT 252). Powdered samples were reacted with dehydrated phosphoric acid under vacuum at 70 °C. The isotope ratio measurement is calibrated based on repeated measurements of NBS-19 and NBS-18 and precision within 1 σ error is $\pm 0.10\text{‰}$ for $\delta^{18}\text{O}$ and $\pm 0.08\text{‰}$ for $\delta^{13}\text{C}$.

3.5. Chronology

Accelerator Mass Spectrometry (AMS) radiocarbon dating analyses were carried out at the Woods Hole NOSAMS. Six radiocarbon dates for Hz11-P03 have been documented by Eriş et al. (submitted) (Table 1). Eriş et al. (submitted) showed the existence of a hiatus by lithological description and seismic stratigraphy at a depth of 270 cm in the core. The radiocarbon dates were calibrated using IntCal (Reimer et al., 2013) and the age model was constructed using the Bacon (Bayesian accumulation) package (Blaauw and Christen, 2011) in the freeware statistical software package, R (R Core Team, 2017).

Unlike “traditional” simple linear age-depth interpolation models, Bacon aims to produce more realistic age models, by applying bivariate monotone Markov process between predefined increments between dated intervals. According to Blaauw and Christen (2011). The qualitative difference between the Bacon and

Table 1

Radiocarbon dates obtained from Hz11-P03. Error intervals, within 95% confidence range, and weighted mean ages are the results of Bacon.

Lab ID	Depth (cm)	Material	^{14}C BP	95% confidence interval (cal BP)	Weighted mean age (cal BP)
OS-106010	59	Mollusc	2240 $\pm$ 45	2012–2345	2209
OS-97568	133	Wood	3560 $\pm$ 45	3717–4327	3939
OS-106006	208	Mollusc	7800 $\pm$ 70	8317–8926	8578
OS-106002	244	Mollusc	9240 $\pm$ 70	10201–10690	10416
OS-96328	285	Mollusc	11400 $\pm$ 50	12931–13437	13212
OS-96329	348	Mollusc	14600 $\pm$ 80	17208–18027	17682

other sophisticated age-depth models, such as OxCal (Bronk Ramsey, 2008) or BChron (Parnell et al., 2008), is that, the results from Bacon are more “environmentally inspired” models.

The Bacon package needs two basic pieces of prior information. The first one is the mean accumulation rate and accumulation shape (a parameter which gives the shape of the accumulation rate probability density function—a gamma distribution). Mean accumulation rate value is suggested by Bacon, prior to the calculations. For Hz11-P03, the package estimated a 50 yr/cm mean accumulation rate. The rate is reasonable for Hz11-P03, since the length of the core is 350 cm and the IntCal13 calibrated result at 348 cm is 17.78 $\pm$ 0.215 cal ka BP. The accumulation shape parameter is left to default of Bacon (that is 1.5), a figure that was arrived at by analysing 152 Holocene lake deposits (Goring et al., 2012). The second piece of prior information required is the section thickness. The code runs over the predefined sections according to the length, which in our case is 5 cm and makes the linear interpolations over sections. Since section thickness parameter should be changed if a smoother curve is needed, we implemented the default parameter for it. Beside these two parameters, it is possible to change the memory strength/memory mean and hiatus depth/hiatus shape. Memory parameters, which construct the character of the sedimentation in the age model, are defined as a beta distribution in Bacon. As stated by Blaauw and Christen (2011), changes in memory parameters do not significantly alter the age model and by changing the parameters we tried to fit the posterior distribution to prior distribution. On the other hand, it is important to note that there is a hiatus at a depth of 270 cm in Hz11-P03. The default hiatus interval length of Bacon is 1000 years, and according to the model this length may change. Since, according to the ages below and above the hiatus, it is somewhere between the uncalibrated ages, 14.6 and 11.4ka ^{14}C BP, we assigned the hiatus to the Younger Dryas (YD) period, formally accepted to be during between 12.85 and 11.65 cal ka BP (Masson-Delmotte et al., 2013). We therefore applied the default value, which is 1000 years, for the duration of the hiatus.

3.6. Statistical analyses

3.6.1. Filtering the data

High resolution μ -XRF data may involve analytical and instrumental noise (Croudace and Rothwell, 2015) but a low-pass filter would minimize these errors. Therefore, in order to minimize noise, we applied a first order Savitzky-Golay filter (SGF) to the Hz11-P03 core μ -XRF data with age control points (Savitzky and Golay, 1964). The SGF fits a polynomial to the data with the desired window length and selects the corresponding point on the polynomial curve and replaces the data point with the selected one. For evenly spaced data a first order SGF is nothing more than a simple moving average. However, for irregularly spaced data, which is more usually encountered in paleoclimate time series, a first order SGF is able to give suitable weights to each point by linear regression in the desired window. In order to fit the data to the age model, after applying the SGF to a five point window length, we resampled the

data at one in every five consecutive data points (Fig. 3). After all the above processes, we applied FastICA algorithm by using a commercial software (MATLAB 8.2), for which the code is freely available online.

3.6.2. Independent component analysis

3.6.2.1. The ICA model. Assume that, $\mathbf{X} = [x_1, x_2, \dots, x_n]$ is a $d \times n$ dimensional observed data matrix and $\mathbf{S} = [s_1, s_2, \dots, s_n]$ is the p -dimensional source random variable, with non-Gaussian and independent entries. If the independent components are mixed by an unknown linear transformation $\mathbf{A}_{d \times p}$ by

$$\mathbf{X} = \mathbf{AS}, \quad (1)$$

then the model in equation (1) is called the ICA model (Hyvärinen and Oja, 2000). To solve the ICA model, the mixing matrix, $\mathbf{A}$, must be known. After estimating the matrix $\mathbf{A}$, $\mathbf{S}$, the source, can be found by

$$\mathbf{S} = \mathbf{WX}, \quad (2)$$

where $\mathbf{W}$ is the inverse of $\mathbf{A}$.

3.6.2.2. Preprocessing. Before applying ICA, the data should be centred and then whitened. Whitening is a process of finding standardized principal components, i.e., principal components with unit variances. This process clears the second-order statistical information of the data. The whitening process of the mixing matrix allows ICA to rotate the axes of the whitened mixed components to the space of independent components, which is essential to liberate the data from the dependency of the covariances (Hyvärinen et al., 2009).

3.6.2.3. The FastICA algorithm. Independent Component Analysis is a method to separate non-Gaussian signals by imposing the probabilistic constraint of minimizing mutual information. It achieves the separation by making use of “Kullback-Leibler divergence”, a measure of difference between two probability distributions of separate signals that become mixed through a variety of processes. As such, unlike the PCA, ICA's starting point is statistical independence which automatically imposes the separability condition on the joint probability distributions of the mixed signals. There are various ways in which ICA can be implemented. In order to extract independent components, Hyvärinen and Oja (2000) offer two different methods. These are (a) the deflation method and (b) the symmetric method. The former method estimates independent components one-by-one, whereas the latter estimates all the components together. Hyvärinen et al. (2004) recommend the deflation method, if the number of independent components is limited. Therefore, the deflation method was preferable for two reasons: firstly to follow the recommendation of Hyvärinen et al. (2004) and secondly, because the distance correlation results of the reference data with the selected independent components (see 3.6.2.4) extracted by the deflation method are higher than the

components extracted by the symmetric method.

The algorithm of the deflation method is described as follows. To estimate only one independent component, Hyvärinen (1999) offers the following algorithm.

- i A non-quadratic contrast function G should be chosen. The contrast function is used to estimate the negentropy (for detailed explanations, see Hyvärinen (1998)), which is a measure of the Gaussianity of a random variable. Negentropy J of a random variable $\mathbf{y}$ is given by

$$J(\mathbf{y}) = H(\mathbf{y}_{\text{gauss}}) - H(\mathbf{y}), \quad (3)$$

where $\mathbf{y}_{\text{gauss}}$ is the Gaussian random variable with the same covariance matrix of $\mathbf{y}$ and H is the measure of Shannon's entropy, which is:

$$H(\mathbf{y}) = -\sum_i p(y_i) \log p(y_i). \quad (4)$$

Note that, negentropy of a random variable is zero if and only if it is Gaussian. Maximizing the negentropy of a random variable means to minimize its Gaussian properties and also, maximizing the negentropy of random variables, which have zero mean and unit variance, means minimizing the mutual information, which is a natural measure of independence (Hyvärinen and Oja, 2000).

The contrast function G should be non-quadratic and for this study we used

$$G(u) = \log(\cosh(u)), \quad (5)$$

which is offered by Hyvärinen (1999) for general purposes.

- ii A random weight vector $\mathbf{w}$ should be chosen in order to initialize the procedure.
- iii By applying Kuhn-Tucker conditions to maximize the negentropy, Hyvärinen and Oja (2000) present the following equation:

$$\mathbf{w}^+ = E[\mathbf{x}g(\mathbf{w}^T\mathbf{x})] - E[g'(\mathbf{w}^T\mathbf{x})]\mathbf{w}, \quad (6)$$

where g is the first derivative of the contrast function G , and

$$\mathbf{w} = \mathbf{w}^+ / \|\mathbf{w}^+\| \quad (7)$$

This procedure is then operated iteratively until the convergence to maximal non-Gaussianity is established. Here, the convergence means that, during the iteration the resulting vector $\mathbf{w}$ should almost point in the same direction as the previous one through the iteration process. They call this method the fixed-point algorithm.

To estimate the remaining components, running the algorithm by assumed component number times by different initial random vectors alone is not enough, because these can converge to the same maxima. To prevent this, decorrelation of the components (which means orthogonalization in the whitened space) is sufficient (Hyvärinen, 1999). After estimating the first p component vectors, i.e., $\mathbf{w}_1 \dots \mathbf{w}_p$, the deflation scheme proposed to find the remaining components is

$$\mathbf{w}_{p+1} = \mathbf{w}_{p+1} - \sum_{j=1}^p (\mathbf{w}_{p+1}^T \mathbf{w}_j) \mathbf{w}_j \quad (8)$$

and then normalizing $\mathbf{w}_{p+1}$ by dividing to its vector length.

Discussion of the advantages of the FastICA algorithm beyond

the other independent component algorithms is beyond the scope of this article but they are discussed in detail by Hyvärinen and Oja (2000).

If the number of the variables is more than the sample size, or the variables have strong time dependencies (i.e., the time series are exactly periodic), then ICA algorithm may produce components with exaggerated spikes or bumps (Hyvärinen et al., 1999), this situation is known as "overlearning". In our case, there are 6 variables each with 634 data points and the elemental profiles are not time dependent, which can be tested by spectral analysis or just by looking at the data (see Fig. 4).

3.6.2.4. Selecting the appropriate components. The resulting vectors of ICA, unlike PCA, are neither given in order nor are their energies known (Hyvärinen and Oja, 2000). Therefore, to select the appropriate components, we applied the distance correlation (Székely et al., 2007) as the similarity measure between the independent components and past climate records, which for this study are $\delta^{18}\text{O}$ for NGRIP (North Greenland Ice Sheet Project) (Andersen et al., 2004) and $\delta^{13}\text{C}$ for Sofular Cave (Fleitmann et al., 2009). Polar ice core $\delta^{18}\text{O}$ data carry information about temperature (Joussame and Jouzel, 1993). Furthermore, NGRIP is the latest drilled ice core and amongst the other ice cores, NGRIP $\delta^{18}\text{O}$ data is the most sensitive to Holocene climate changes (Johnsen et al., 2001). The Sofular Cave stable isotope sequences provide high resolution records for the Eastern Mediterranean region, with a more than adequate age model. We will therefore use these datasets as the reference data hereafter.

The empirical distance correlation for ordered pair data, $(\mathbf{x}, \mathbf{y})$ with n elements, is defined as follows:

$$R(\mathbf{x}, \mathbf{y}) = \begin{cases} \sqrt{\frac{\nu^2(\mathbf{x}, \mathbf{y})}{\sqrt{\nu^2(\mathbf{x})\nu^2(\mathbf{y})}}} & , \text{ if } \nu^2(\mathbf{x})\nu^2(\mathbf{y}) > 0; \\ 0 & , \text{ if } \nu^2(\mathbf{x})\nu^2(\mathbf{y}) = 0, \end{cases} \quad (9)$$

where

$$\nu^2(\mathbf{x}, \mathbf{y}) = \frac{1}{n^2} \sum_{k,l=1}^n A_{kl} B_{kl}, \quad (10)$$

and

$$\nu^2(\mathbf{x}) = \frac{1}{n^2} \sum_{k,l=1}^n A_{kl}^2. \quad (11)$$

A_{kl} and B_{kl} are defined as

$$A_{kl} = a_{kl} - \bar{a}_k - \bar{a}_l + \bar{a}_{..} \quad (12)$$

and

$$B_{kl} = b_{kl} - \bar{b}_k - \bar{b}_l + \bar{b}_{..}, \quad (13)$$

where $a_{kl} = \|X_k - X_l\|$, $b_{kl} = \|Y_k - Y_l\|$ and $k, l = 1, \dots, n$. The subscript $..$ denotes that the mean is computed for the whole index.

The distance correlation is equal to 0 if and only if the random variables are independent and $0 \leq R(\mathbf{x}, \mathbf{y}) \leq 1$. The distance correlation's superiority comes from its power to catch non-linearities, strength on outliers, revealing independencies and computational simplicity. However, its statistical power over rival methods is an on-going topic of discussion (Kinney and Atwal, 2014; Martínez-Gómez et al., 2014; Richards et al., 2014; Simon and Tibshirani,

Table 2

Distance correlation coefficients of HZ11-P03 independent components of μ -XRF data with NGRIP $\delta^{18}\text{O}$, Sofular $\delta^{13}\text{C}$ and μ -XRF data.

dcorr	HZ-ic1	HZ-ic2	HZ-ic3	HZ-ic4	HZ-ic5	HZ-ic6
NGRIP- $\delta^{18}\text{O}$	0.16	0.17	0.17	0.38	0.47	0.43
Sof- $\delta^{13}\text{C}$	0.13	0.15	0.15	0.26	0.43	0.27
Ca	0.21	0.51	0.14	0.38	0.48	0.53
Fe	0.17	0.23	0.20	0.54	0.57	0.54
K	0.16	0.24	0.23	0.58	0.18	0.77
Mn	0.57	0.28	0.22	0.22	0.62	0.44
Sr	0.18	0.45	0.16	0.77	0.29	0.29
Ti	0.12	0.34	0.52	0.76	0.19	0.23

2014).

Distance correlation, as inputs, needs data vectors each having the same number of elements. Therefore we linearly interpolated the NGRIP $\delta^{18}\text{O}$ record, with 20 years of resolution (Andersen et al., 2007), and the Sofular Cave $\delta^{13}\text{C}$ record to HZ11-P03 age model and computed the distance correlation coefficients to identify any climate-related components (Table 2).

4. Results and discussions

According to the age model, HZ11-P03 covers the last 17.3 cal ka BP with a hiatus of 700 years between 12.49 cal ka and 11.76 cal ka BP (Fig. 2). The age of the hiatus, according to the dates calculated by using Bacon, fit almost within the YD cold event. This estimate of the hiatus fits with the hypothesis proposed by Eriş et al. (submitted) of a 73 m drop in the level of Lake Hazar during the YD period.

The idea of decomposing mixed data by means of statistical independence has some superiority over orthogonal decomposing

with respect to maximal variance (Whitfield et al., 1999; Westra et al., 2007). However, there are some drawbacks. The first one is the assumption of non-Gaussianity of the source signal. The non-Gaussianity of modern climate variables (Perron and Sura, 2012; Sura and Hannachi, 2014) makes this assumption plausible. Secondly, the order of the resultant independent components and their signs are unknown. While the assumptions of ICA come from a stronger assertion than the second order statistical techniques, it is not easy to select and give meanings to the results. Therefore, in this study, the candidate components are selected through a similarity measure, which is the distance correlation. Components that are highly correlated with the well dated, high resolution records are taken as basis and the correlation results with the μ -XRF data are discussed. However, especially climatologically well established periods, such as the Bølling-Allerød, make it easy to select the desired components with the correct sign. Here, the use of reference data is not to create a circularity, rather it is used to have a basis on selecting the appropriate components.

The other drawback of the method lies in the nature of the μ -XRF data. It is clear that ICA cannot find any result which does not exist within the data. The values of local extremes in the data may not only depend on natural factors, but also on the precision of the ITRAX core scanner. For example, a period with higher amount of coarse grained sediment input may distract the μ -XRF results (MacLachlan et al., 2015). So, while evaluating the magnitude of ICA of μ -XRF data results, it should be kept in mind that the results may be temporally local. It would be better to evaluate a period with events before and after.

A question may arise about the data analysed. One may want to apply the method on elemental ratios which is widely used in geochemical proxy data interpretations. We also applied FastICA on the Fe, K, Ti, Ca/Ti, Sr/Ca and Mn/Fe dataset, however the results of both analyses are convincingly similar.

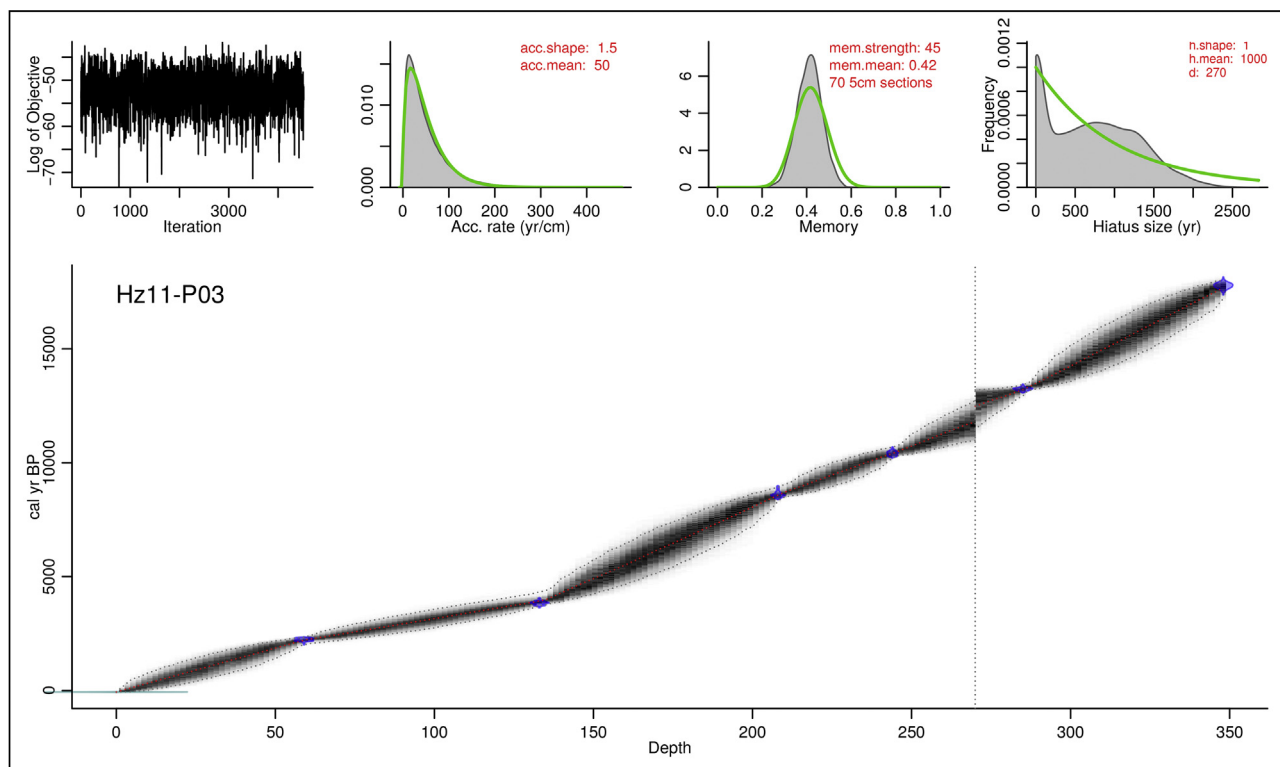


Fig. 2. Age model of HZ11-P03, constructed using the Bacon statistical package. The dashed line at 270 cm shows the hiatus location.

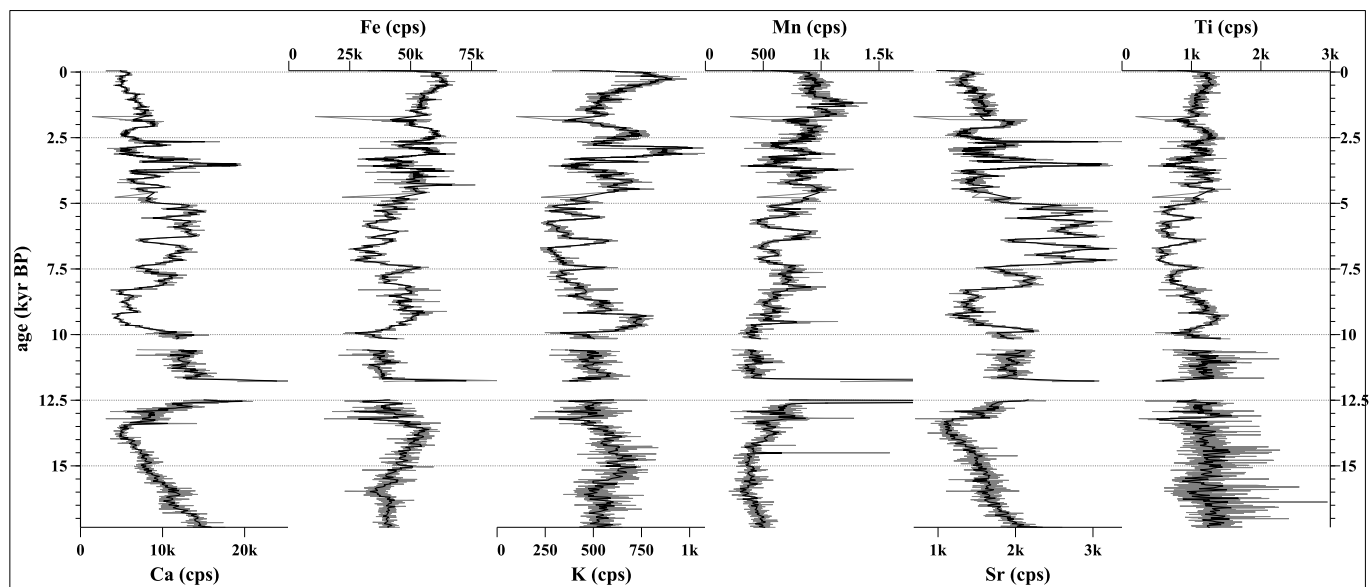


Fig. 3. The μ -XRF data in gray and on top of each profile the filtered data in black as described in 3.6.1 are plotted.

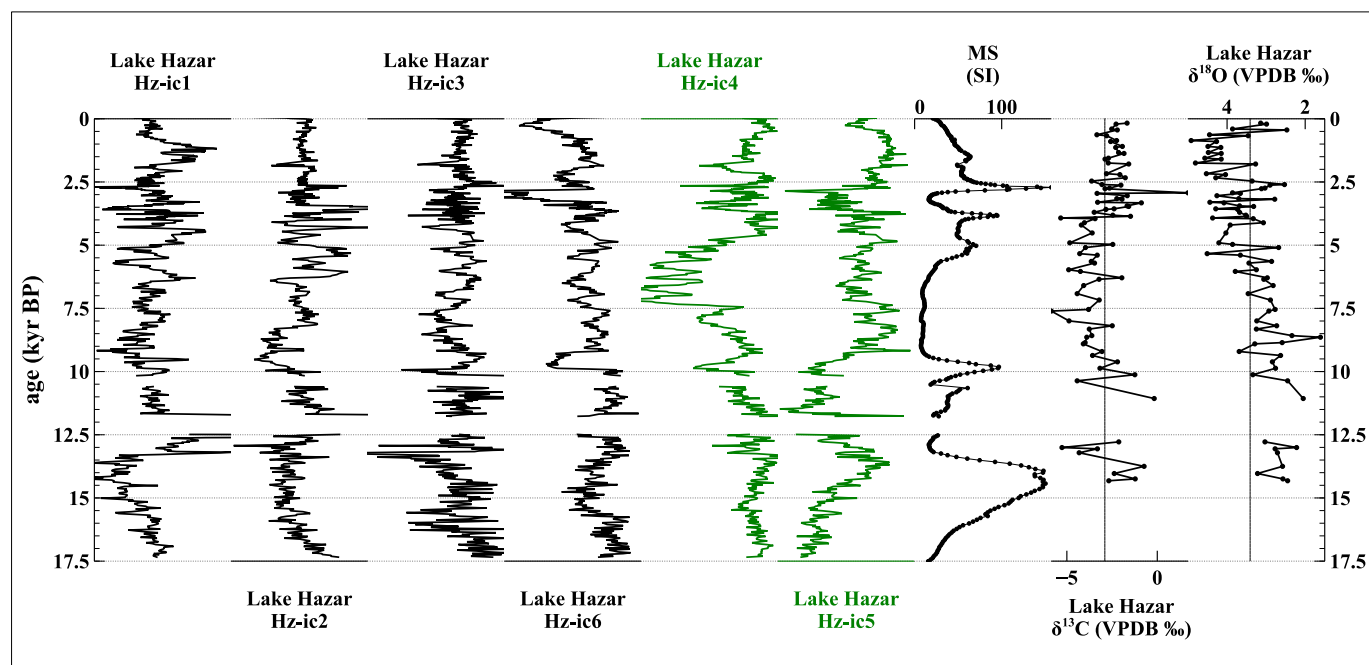


Fig. 4. Components after the FastICA, MS and stable isotope data. The vertical lines at the isotope plots show the average of isotope data. Green colored plots (Hz-ic4 and Hz-ic5) are the selected independent components. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.1. Interpretation of the independent components

The μ -XRF Ca, Fe, K, Mn, Sr and Ti concentrations that are extensively used in paleolimnology studies (Davies et al., 2015), constitute around 95% of the total measured content of the μ -XRF data of the Hz11-P03 core. We therefore applied FastICA to the profiles of these six elements and the resulting components were compared with the reference data and the μ -XRF data via distance correlation.

The distance correlation coefficients between the independent components and the reference NGRIP $\delta^{18}\text{O}$ and Sofular $\delta^{13}\text{C}$ data

show that the fifth independent component (Hz-ic5) has the highest correlation (Table 2). Moreover, the correlation coefficients of independent components with the μ -XRF data imply that Hz-ic5 is also related to Ca, Fe and Mn (Table 2). Ca may precipitate as calcium carbonate in lakes via a number of different processes (cf. Kelts and Hsü (1978); Çağatay et al. (2014)). Solubility of carbonate minerals is dependent on temperature and salinity. Their precipitation may be related to the organic productivity and evaporation processes, which indirectly related to temperature, evaporation/precipitation and the availability of light. Fe and Mn can be used as a measure of detrital input but both also contain information about

reducing conditions in lakes (for extensive summaries see Cohen (2003) and Davies et al. (2015)). The redox reactions are mainly determined by organic matter production and water column stratification, which in turn is mostly affected by light and temperature (Gaillardet et al., 2014), and which is commonly coupled with Ca precipitation.

NGRIP $\delta^{18}\text{O}$ reflects the temperature changes in Northern Hemisphere high latitudes (Joussaume and Jouzel, 1993) and according to Fleitmann et al. (2009), the changes in temperature are reflected in the $\delta^{13}\text{C}$ record from Sofular Cave stalagmites. Therefore, the correlation of Hz-ic5 with the reference data sets and Ca, Fe and Mn could be argued to reflect temperature changes in the lake and its catchment region (Fig. 5).

While the sixth component (Hz-ic6) has the second highest correlation with the reference data, we could not attach any meaning to it. However, the high correlation values of the fourth component (Hz-ic4) with Ti, Sr and also with the reference data (Table 2), give an indication about a relationship between Hz-ic4 and precipitation. It correlates closely with Ti, Sr, K and Fe (Table 2). Ti, K and Fe provide information about detrital input that is mainly dependent on precipitation. Sr is used as a proxy for evaporative processes in lakes (Cohen, 2003). Thus, we can interpret Hz-ic4 as a proxy for precipitation/evaporation (Fig. 5).

Summarizing our results, we conclude that Hz-ic4 and Hz-ic5 are proxies of precipitation/evaporation and temperature, respectively. However, it should be noted that they have no absolute units, only reflecting relative changes through time, and are referred as "temperature" and "precipitation" for the rest of this paper.

4.2. Temperature and precipitation records

The results from Lake Hazar reveal that before 14.6 ka BP the temperatures of the region were lower than the following warm period. On the other hand, the precipitation was at almost the same level as the subsequent Bølling-Allerød warm period. Temperatures in the region rise with the start of the Bølling-Allerød and the precipitation record shows a slight increase with this warming. At

the end of Bølling-Allerød, the lake level evidently dropped to 73 m below its modern level (Eriş et al., submitted) producing a 700 yr hiatus during the YD period.

After the start of the Holocene, temperatures evidently rose gradually and achieve the warm conditions of the Holocene within 2 millennia. After 9.5 ka BP, Hz11-P03 temperature record follows the North Atlantic stacked ocean record (Bond et al., 2001) within the error range of Hz11-P03 age model. The most striking of all the oscillations is the one around 3.5 ka BP, which corresponds to the second Bond cycle (Bond et al., 2001). This cold period (Table 2) lasted until 2.6 ka BP.

The precipitation record for the Holocene is somewhat different to the temperature record. The start of the Holocene was as wet as the Bølling-Allerød interstadial and but a gradual decrease in precipitation occurred up to 9.5 ka BP. From 9.5 ka BP, until 8 ka BP, the region experienced a wet period. After 8 ka BP, and lasting for almost 3 ka, a harsh dry period with internal oscillations can be observed. This arid phase is also evident in MS and $\delta^{13}\text{C}$ records of Hz11-P03. After 5 ka BP a wet phase occurred following almost directly on from the oscillations in the temperature record, with marked dry phases taking place during 3.5 ka; 2.8 ka; 1.8 ka BP (Fig. 5).

4.3. Comparison with other regional records

4.3.1. Pre-holocene

Pollen (Prentice et al., 1992; Mudie et al., 2002), lake level (Roberts and Wright, 1993; Erol, 1997) and glacier (Sarıkaya et al., 2008, 2014) studies from around the Eastern Mediterranean indicate that the region was cold, but humid after the Last Glacial Maximum (LGM) until the Bølling-Allerød interstadial. In this context, the Lake Hazar results, which indicate, relatively cold and humid conditions before the Bølling-Allerød, correlate well with these other regional records. There exist two main hypotheses about the cause of these cold but wet conditions in the Eastern Mediterranean between the LGM and the Bølling-Allerød. The first is that cool and cloudy summers, that decrease evaporation, are the

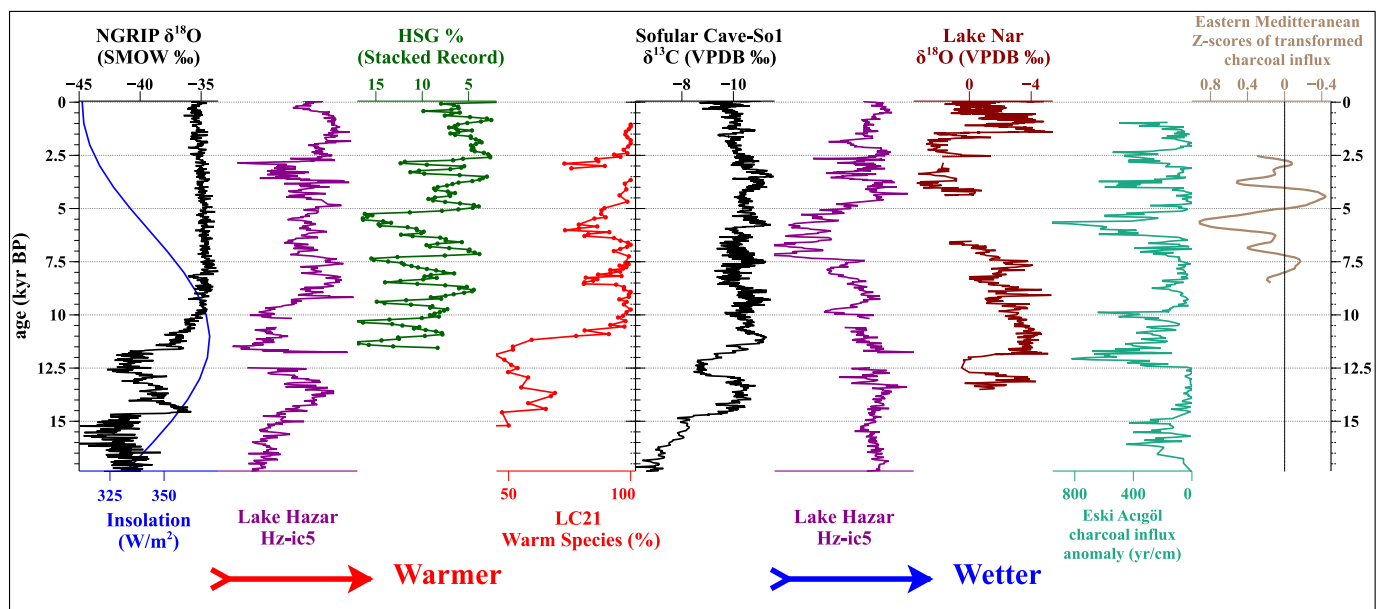


Fig. 5. Comparison of the results of this study with other studies. From left to right, NGRIP $\delta^{18}\text{O}$ (Andersen et al., 2007) and difference between summer solstice and winter solstice insolation for 37°N (Berger and Loutre, 1991), Hz-ic5, North Atlantic hematite stained grains percent (Bond et al., 2001), percent of the LC21 core warm species (Rohling et al., 2002), Sofular Cave $\delta^{13}\text{C}$ (Fleitmann et al., 2009), Hz-ic4, Lake Nar $\delta^{18}\text{O}$, as explained in the original study, 3% shifted Eski Acıgöl charcoal influx anomaly (Turner et al., 2008), average age transformed charcoal influx for Eastern Mediterranean lakes (Vannière et al., 2011).

reason for semi-arid vegetation and high lake levels in this interval (COHMAP Members, 1988). The second hypothesis is that the cause was the southward migration of the jet streams, and resultant moisture-rich westerly winds during LGM (Prentice et al., 1992; Harrison et al., 1996). Such a migration would cause wetter conditions in the Eastern Mediterranean. The precipitation result from Lake Hazar for the 17.3 and 14.8 ka BP interval, support the migration of the westerlies hypothesis, although we cannot entirely reject the cool and cloudy summer hypothesis. Probably, both mechanisms were in effect in the Eastern Mediterranean before the Bølling-Allerød.

There are studies in the Eastern Mediterranean which offer a dry period during the Heinrich-1 event (Bar-Matthews et al., 1999; Fontugne et al., 1999). However, our H11-P03 data and ICA results do not offer an arid phase for ~17 ka BP.

Following this, Bølling-Allerød period was warm and wet by comparison, as one would expect based on current models and data. As previously documented by Eriş et al. (submitted), the decrease in lake level and resulting hiatus within the core between 12.49 and 11.76 cal ka BP shows that, for the last 17.3 ka, the region experienced its most extreme dry period during the YD. Also, the low temperature values at the end of the Bølling-Allerød period and at the start of the Holocene indicate that the YD period is likely to have been as cold as expected.

4.3.2. The Holocene

From the end of the YD period until 9.5 ka BP, the temperature of the region shows an increasing trend, as seen in the NGRIP $\delta^{18}\text{O}$ record in the same period. The oscillations in temperature resemble those seen in the North Atlantic stacked HSG record (Bond et al., 2001). However, the precipitation record behaves somewhat differently to that for temperature. For the early Holocene, the Lake Nar $\delta^{18}\text{O}$ record (Dean et al., 2015) and H11-P03 precipitation record closely resemble each other with a wet period up until 9.5 ka BP, and a decreasing precipitation trend to 8.2 ka BP. The wet conditions with a drying trend were also documented in previous Eastern Mediterranean studies (Harrison et al., 1996; Bar-Matthews et al., 1997; Roberts et al., 2001). During the early Holocene, summer insolation was at its maximum, whereas winter insolation was at its minimum (Fig. 5). This increasing seasonality may have resulted in the Mediterranean Sea being warmer during cold winters and consequently, as Arz et al. (2003) have proposed, it was wetter at the start of the Holocene.

After 8 ka BP, while the temperature oscillations follow the North Atlantic climatic oscillations (Bond et al., 2001), precipitation records show a period of great aridity until 5 ka BP. During this period, North Africa was wetter (deMenocal et al., 2000) and evidence for a drought in the Eastern Mediterranean is indicated by charcoal data (Turner et al., 2008; Vannière et al., 2011). During this interval, as the Northern Hemisphere was receiving more insolation than today, the Intertropical Convergence Zone lay in a more northerly position (Haug et al., 2001; Arbuszewski et al., 2013). As a result of this, the subtropical high pressure belt, which today lies over the southern Mediterranean, particularly affected the north-east Mediterranean and created arid conditions in the region.

Between 5.5 and 5 ka BP the precipitation record reveals that the region gradually became wetter, finally reaching modern conditions. However, the abrupt decline in both precipitation and temperature around 3.5 ka, 2.8 ka and 1.8 ka BP, which were also documented in the seismic records (Eriş et al., submitted), may have been the result of a coincidence of the strengthening of the Siberian high pressure system during winters (Rohling et al., 2002; Çağatay et al., 2014), and the gradual decrease in solar irradiance, especially around 2.8 ka BP (Roth and Joos, 2013), in accordance

with changes in the North Atlantic (Bond et al., 2001).

For the Holocene, the most striking result is that the spikes in precipitation and temperature records appear to closely follow the North Atlantic Bond events, whereas the trends do not. If the cause of the Bond events is indeed solar forcing, as claimed by Bond et al. (2001), then we can also state that the climate oscillations in the region were also greatly influenced by solar forcing. On the other hand, if the cause were related to circulation changes in the North Atlantic (Debret et al., 2007), then it is noteworthy to add that there appears to be a direct relationship between the temperature of the region and North Atlantic. Additionally, we can state that precipitation oscillations are related to the North Atlantic, but that precipitation appears to be a function of a much more complex system.

The regional context of Lake Hazar is such that not only do the results of the current study have implications for understanding the Holocene climate and environment of eastern Anatolia but also for that of Mesopotamia. This is because, when at its maximum, the waters that flow out from Lake Hazar empty into one of the two great Mesopotamian rivers, the Tigris. Together with the Euphrates, it is rainfall and meltwater from the high mountains of northern Mesopotamia and eastern Anatolia that feed the Tigris and formed the economic basis of that region's ancient civilizations, which relied on river-fed irrigation systems from these two rivers as they flowed through an otherwise dry landscape.

A number of scholars have hypothesized that several incidents of societal collapse in Mesopotamia that are observable in the archaeological record may have been caused by the type of abrupt environmental change that we see in the data from Lake Hazar. Although much more research on the question of the temporal coincidence between climate and changes in human settlement is required, it is interesting to make some preliminary observations on how our scientifically observed climate data may relate to these archaeological phenomena.

For example, an observed archaeological phenomenon for which a climatic cause has been posited is the so-called 4.2 ka event (Weiss et al., 1993; Dalfes et al., 1997). However, here the ICA of the H11-P03 core indicates that there is no evidence for any abrupt climatic change at this point in time. We therefore suggest that further studies should be conducted to confirm or deny the existence of abrupt climate change in the Anatolia/Mesopotamia region at this time, but based on our evidence we suggest that an archaeological explanation is more likely to account for this observed cultural phenomenon than a climatic one.

4.4. Comparison with other proxy records of H11-P03

Stable isotope and MS data have average age resolution of 175 yr over the last 14.3 cal ka BP and 51 yr over the last 17.5 cal ka BP, respectively (Fig. 4).

Ostracod shells closely reflect the isotopic-equilibrium conditions of the host water at the time they die (Holmes, 1996). However, oxygen isotope data from Mediterranean lakes are not explicit proxies for climate and need careful interpretation (Jones and Roberts, 2008; Reuter et al., 2017). Roberts et al. (2008) state that, the $\delta^{18}\text{O}$ records of Mediterranean lakes may be the product of some independent factors, such as changes in temperature, seasonality, air mass trajectory or precipitation/evaporation modified by groundwater influx. Hence, $\delta^{18}\text{O}$ data is not a climate proxy of a single independent process. On the other hand, the $\delta^{18}\text{O}$ data from Lake Hazar has an increasing trend from the Bølling-Allerød to the end of the Holocene, except for the last 500 years when the $\delta^{18}\text{O}$ values decrease.

Stable carbon isotopic fractionation in closed lakes is governed by evaporative processes and dissolution of carbonate grains, but mainly by biotic/abiotic interactions (Kelts and Talbot, 1990). The

Lake Hazar $\delta^{13}\text{C}$ record, especially during 9 and 5 ka BP, resembles the precipitation record, with light $\delta^{13}\text{C}$ values of ostracod shells corresponding to low precipitation, most probably related to photosynthetic response to drought stress of the plants in and around the lake (Fig. 4). Furthermore, MS measurement, which is a reflectance of the mineral composition and therefore dependent on depositional conditions, closely follows the precipitation and $\delta^{13}\text{C}$ record, especially through the mid-Holocene.

5. Conclusions

In this paper, we present a new method to extract paleoclimate signals from μ -XRF data using independent component analysis (ICA) and the stable isotope data from Lake Hazar.

ICA proves to be more suited to the extraction of paleo-climate signals than the traditional second order statistical methods (such as PCA and FA), if it is known that the underlying signals are the results of statistically independent/quasi-independent processes. ICA's aim is to find directions in space that satisfy maximum independence rather than to find directions with maximal variance. In this study, the aim is to offer a method for geochemical time series, which are thought to be representatives of past climates. However, it should be noted that, particularly, the process of component selection from the results of ICA (see section 3.6.2.4) needs to be improved for future studies.

The μ -XRF data from the Hz11-P03 sediment core from Lake Hazar shows six different elemental profiles, which constitute around 95% of the total count, which were subjected to ICA, and accordingly six independent components were gathered. We selected two of the six independent components from the Lake Hazar μ -XRF data, based on their distance correlations against NGRIP and Sofular Cave isotope data. Our paleoclimatic and paleoenvironmental interpretations of these two components suggest that they are related to precipitation/evaporation and temperature. Based mainly on the profiles of these two independent components, we conclude the following:

1. The interval between 17.3 and 14.8 ka BP was cold but as wet as the following the Bølling-Allerød period, a conclusion which is also supported by previous studies around the Eastern Mediterranean. This was, most probably, the result of the southward migration of the Northern Hemisphere jet streams which were restrained from moving northwards by the anti-cyclonic atmospheric system settled over the high latitude LGM ice sheets.
2. The Bølling-Allerød warm interval, which is dated to between 14.8 and 12.85 ka BP, was warm and wet.
3. During the Younger Dryas cold event, the level of Lake Hazar fell by 73 m, causing a hiatus in Hz11-P03 during 12.49 and 11.76 ka BP.
4. At the start of the Holocene temperatures rose gradually, and reached "Holocene normals" at around 8 ka BP. This period after the YD was unexpectedly wet because of seasonality due to orbital parameters.
5. Between 8 and 5 ka BP, the region's temperature record follows the North Atlantic oscillations. However, its precipitation record shows that it experienced its most arid period during the Holocene. This was the result of the settling of the Subtropical High Pressure system over the north-eastern Mediterranean which blocked humid air streams from the Atlantic. In accordance with the changes in insolation, this dry period gradually came to an end around 5 ka BP.
6. After 5 ka BP, and up until 3.5 ka BP the climate again fluctuated around the "Holocene normal". However, around 3.5 cal ka BP, temperatures fell abruptly and sudden cold arid phases occurred around 3.5 ka, 2.8 ka and 1.8 ka BP. The cause of this abrupt

cooling and drying phase can be the strengthening of the Siberian high and in accordance with the changes in insolation and North Atlantic system.

7. There is no evidence for the putative 4.2 ka climate event in Lake Hazar which has been postulated on the basis of archaeological evidence.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://doi.org/10.1016/j.quaint.2017.08.066>.

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